

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 40

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• t-test • The data are measured (continuous) values and two means are being compared • The data are (assumed to be) normally distributed	3
01.2	• Use the Mann-Whitney U test instead (compares medians)	1
01.3	• There is a 2% probability that the difference occurred by random chance • As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected	2
01.4	• The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 • About 68% of values lie within one SD of each mean so the two data sets are clearly separated	2
01.5	• Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists • Another variable (eg topography / aspect of the sites) could cause both; a controlled experiment or further evidence is needed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
02.1	• Conventional: $40 / 100 = 40\%$ • CHP: $(35 + 45) / 100 = 80\%$	2
02.2	• Extracting less energy as electricity leaves the cooling / condenser water at a higher temperature • Hotter water is more useful for district heating of homes, fish farms, greenhouses	2
02.3	• Hot water loses heat and needs pumping energy over distance; heat cannot be transported as far as electricity	1

Question	Marking guidance	Marks
02.4	- Long narrow pipes give a large surface area and copper is a good thermal conductor so heat transfers quickly - Hot and cold fluids flowing in opposite directions maintain a temperature gradient along the whole length so more heat is transferred - Slower flow allows more time for heat to transfer	3
02.5	- One large tank has a lower surface area : volume ratio than many small ones - A sphere has the smallest surface area for any volume; less surface means less heat loss (with insulation of low thermal conductivity)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 233-235; AQA spec 3.3.4.2.3.	
03.1	- Renewable: re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel) - Depletable: use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably - Wood / biofuels are renewable but depletable if forests are felled faster than they regrow	3
03.2	- Other factors (technology, location, cost) restrict how much can be harnessed - Examples: coal too deep to mine; winds high above the ground / far offshore where turbines cannot be placed; waves in deep water where equipment cannot be anchored; low-intensity sunlight that cannot produce high temperatures - New technologies or higher prices may make these usable in future	3
03.3	<i>Any 4 from:</i> - Wind: strong reliable winds - shallow seas, open plains, uplands; low land-use conflict (not near urban areas, ecologically sensitive or scenic areas); near the grid - HEP: high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock - Solar: high light intensity, low cloud cover - Wave: coasts with strong reliable prevailing winds and a long fetch - Tidal: large tidal range, coastal features (estuaries, channels) that focus the flow - Geothermal: hot rocks near the surface / recent volcanic activity - Biofuels: nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.	
04.1	$(4.8 - 1.4) / 4.8 \times 100$ = 70.8% (accept 71%)	2
04.2	- A tributary joins the river, more than doubling the flow (15 to $36 \text{ m}^3 \text{ s}^{-1}$) - So the pollutant is diluted by the extra water (dispersal), not broken down	2
04.3	- The pollutant is persistent so it is not degraded / broken down as it travels - Little extra water joins the river (flow 36 to 38) so little further dilution	2

Question	Marking guidance	Marks
04.4	<i>Any 2 from:</i> • Lake has a long residence time / water does not flow away so the pollutant accumulates rather than being carried away • Smaller volume of water / less dilution / less dispersal by currents • Persistent pollutant gradually becomes more concentrated (eg by evaporation)	2
04.5	• Marine discharge: large volume of water and currents dilute and disperse the pollutant to acceptably low concentrations (control of emission location) • River: small volume so concentrations stay high / drinking water abstraction / sensitive wildlife downstream	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-241, 254-255; AQA spec 3.4.1, 3.4.2.1.	
05.1	• High mean wind velocities that are consistent in strength and direction • A long fetch (distance of open water) over which waves build up, eg Atlantic coasts of the UK	2
05.2	• Oscillating water column (eg Islay Limpet): waves force water up and down in a submerged chamber; the air forced in and out drives a turbine • Or point absorber (floating buoy moving relative to a fixed base turns a generator); surface attenuator (Pelamis - hinged sections push pistons pumping fluid over a turbine); overtopping device (Wave Dragon - waves fill a reservoir above sea level which drains through turbines); oscillating wave surge converter (Oyster)	2
05.3	<i>Any 2 from:</i> • Equipment must withstand storms and corrosion • Difficult to anchor off stormy coasts / in deep water • Expensive to transmit electricity from isolated coasts to consumers • Competition with established technologies	2
05.4	• Barrage: huge quantities of construction materials; reduces tidal range so intertidal mudflats used by wading birds are lost; changes sedimentation (erosion near turbines, deposition elsewhere), lower turbidity; pollutants concentrate in the lagoon; blocks migratory fish and shipping • Lagoon: similar but smaller scale; only part of the estuary enclosed so the most sensitive areas and main channel can be avoided • In-stream turbines: do not block the tidal flow so little impact on range or sediment; but noise affects cetaceans and many dispersed turbines could have a larger total impact; harness less energy	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 210-211, 214-217; AQA spec 3.3.4.1.	
06.1	• $25 + 5 = 30\%$	1

Question	Marking guidance	Marks
06.2	• UV: short wavelength; mostly absorbed in the stratosphere by monatomic, diatomic and triatomic oxygen (ozone layer) which are formed and destroyed in a dynamic equilibrium; converted to chemical energy in these reactions; little reaches the surface • Visible light: transmitted through the atmosphere easily (some reflected by clouds / dust); absorbed by the Earth's surface and converted to heat; some absorbed by chlorophyll and converted to chemical energy in photosynthesis • IR: emitted by the warm surface (long-wave); absorbed by greenhouse gases (CO₂, water vapour, methane) and converted to heat; the warm atmosphere re-emits IR to the surface and reduces conduction losses; some escapes to space	6
06.3	• Greenhouse gases absorb most of the IR emitted by the surface and re-radiate it in all directions, including back down • The surface is repeatedly warmed and re-emits, so the total IR exchanged between surface and atmosphere exceeds the incoming solar energy; only the net loss (70%) balances the absorbed solar energy at equilibrium	2
06.4	• The atmosphere forms layers (troposphere, stratosphere, mesosphere) with different temperature gradients because energy is absorbed at different altitudes (eg UV heating the stratosphere, surface heating the troposphere)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
07.1	• Trade winds blow westwards across the Pacific near the equator, driving a surface current in the same direction • Surface water moving away from South America is replaced by deep, cold water drawn upwards (upwelling) near the coast • The upwelling water is rich in nutrients (nitrates, phosphates) so large algal / phytoplankton blooms occur • These support a rich food web including the anchovy fishery; the water warms as it travels west	4
07.2	• The trade winds weaken or reverse so the surface current slows / reverses and warm water spreads east • Upwelling of nutrient-rich cold water off South America stops • Phytoplankton productivity collapses so the food web / anchovy population collapses; catch fell from about 7.7 to 1.7 million tonnes	3
07.3	<i>Any 2 from:</i> • Heavy rain and floods in the normally dry coastal deserts of Peru / South America • Drought and reduced rainfall in eastern Australia / Indonesia • Droughts in NE Africa, southern Africa, China • Fewer hurricanes in the North Atlantic / fewer cyclones in Japan	2
07.4	• Events appear to be occurring more frequently and are driven by wind patterns that warming may alter, but the causes of ENSO are not fully understood and natural fluctuations make trends hard to separate	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-99; AQA spec 3.2.1.2, 3.2.2.3.	
08.1	• Aerobic decomposition produces little methane and returns nutrients to the soil	1

Question	Marking guidance	Marks
08.2	• It may contain concentrated heavy metals / toxic residues • It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble	2
08.3	<i>Any 2 from:</i> • Impermeable polymer / clay liner to stop leachate escaping • Collection and treatment of leachate • Collection of methane and use as fuel • Daily covering of waste / final capping and restoration	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	
09.1	• Human = $120 + 60 + 30 = 210$; total = 325 • $210 / 325 \times 100 = 64.6\%$ (accept 65%)	2
09.2	• Nitrogen and hydrogen react to form ammonia over an iron catalyst at high temperature and pressure; ammonia then converted to nitrates for fertiliser • Uses large amounts of energy from fossil fuels (also the hydrogen source), releasing CO₂ and NO_x • Biological fixation by bacteria occurs at low temperature using solar energy captured by plants (enzymes lower activation energy)	3
09.3	<i>Any 3 from:</i> • Eutrophication of water bodies by leached nitrates / sewage • Photochemical smogs and tropospheric ozone from NO_x • Acid rain (nitric acid) • Nitrous oxide as a greenhouse gas • NO_x toxicity / respiratory disease • Nitrate in drinking water (methaemoglobinaemia)	3
09.4	• Carbon is lost as CO₂ (composting) or CH₄ (digestion) so the C:N ratio improves / more nitrogen-rich • Bulk and odour reduced, pathogens killed, easier to apply; digestion also yields biogas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 154-155; AQA spec 3.2.4.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Smoke: solid particulates (PM₁₀, PM₅, PM₁) from incomplete combustion; small particles stay suspended, penetrate lungs; toxic chemicals (acids, phenols, metals) carried on particles; high albedo cools climate; blocks light reducing photosynthesis; damages buildings • Smoke smogs form under temperature inversions in valleys (London 1952, 12 000 deaths); synergism with SO₂ • Controls: Clean Air Act 1956 (smokeless zones), smokeless fuels / gas / electricity, coal treatment, electrostatic precipitators, cyclones, bag filters, scrubbers, DPFs, better combustion; smoke smogs have largely been eliminated in UK cities • Acid rain gases: SO₂ (coal combustion, smelting sulfide ores) oxidised by ozone to SO₃ - sulfuric acid; NO_x (high temperature combustion) - nitric acid; HCl from PVC incineration; reactive gases travel long distances (transboundary pollution) • Effects: pH denatures proteins / enzymes in exposed cells (gills, eggs, root hairs, stomata); dissolves exoskeletons; leaches Ca and Mg then mobilises toxic Al and Pb; corrodes metals, dissolves limestone; severity depends on bedrock (limestone neutralises) • Controls: fuel desulfurisation (coal washing, hydrodesulfurisation, low-sulfur ship fuels), dry and wet FGD, fluidised bed combustion, catalytic converters, urea sprays; switching from coal to gas / renewables • Evaluation: SO₂ emissions in Europe have fallen dramatically and lake pH is recovering; NO_x harder to control because of growing vehicle numbers; catalytic converters need lead-free fuel and warm engines; FGD is costly and produces wastes / gypsum; controlling primary pollutants also prevents secondary pollutants (ozone, PANs) • Judgement on relative success of controls for particulates vs SO₂ vs NO_x	9
10.2	<i>Any 1 from:</i> • Reduces growth rates of trees / crops • Damages leaf cuticles (allowing SO₂ to cause more damage)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	

Question	Marking guidance	Marks
11.1	Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Problematic properties of solar: intermittency, unreliability from cloud, low energy density, changing angle; locational constraints (deserts, latitude) • Harnessing solar: photothermal, thermal stores, passive architecture, heat pumps, PV principles and cell efficiencies, solar farms; environmental impacts of manufacture and land use • New solar technologies: multijunction, anti-reflective surfaces, CSP with molten salt, PV-thermal hybrids, transparent PV, heliostats, self-cleaning • HEP: gravitational potential energy, locational constraints (catchment, rainfall, turbidity, impermeable rock, seismic, topography, land use, distance), impacts on habitats, sediment, methane, communities • Energy storage and matching supply to demand: pumped storage, batteries, grid links; combining resources • Evaluation and justified conclusion on how far these resources can replace fossil fuels Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-203, 217-224; AQA spec 3.3.4.1.	25

Question	Marking guidance	Marks
11.2	Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry) • Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips) • Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	25